

AI-POWERED DOCUMENT INTELLIGENCE ASSISTANT

A PRACTICAL RAG SYSTEM FOR PUBLIC-SECTOR PDFS

• Upload • OCR • Summarize • Ask • Cite sources

PITB AI Internship | Final Assignment

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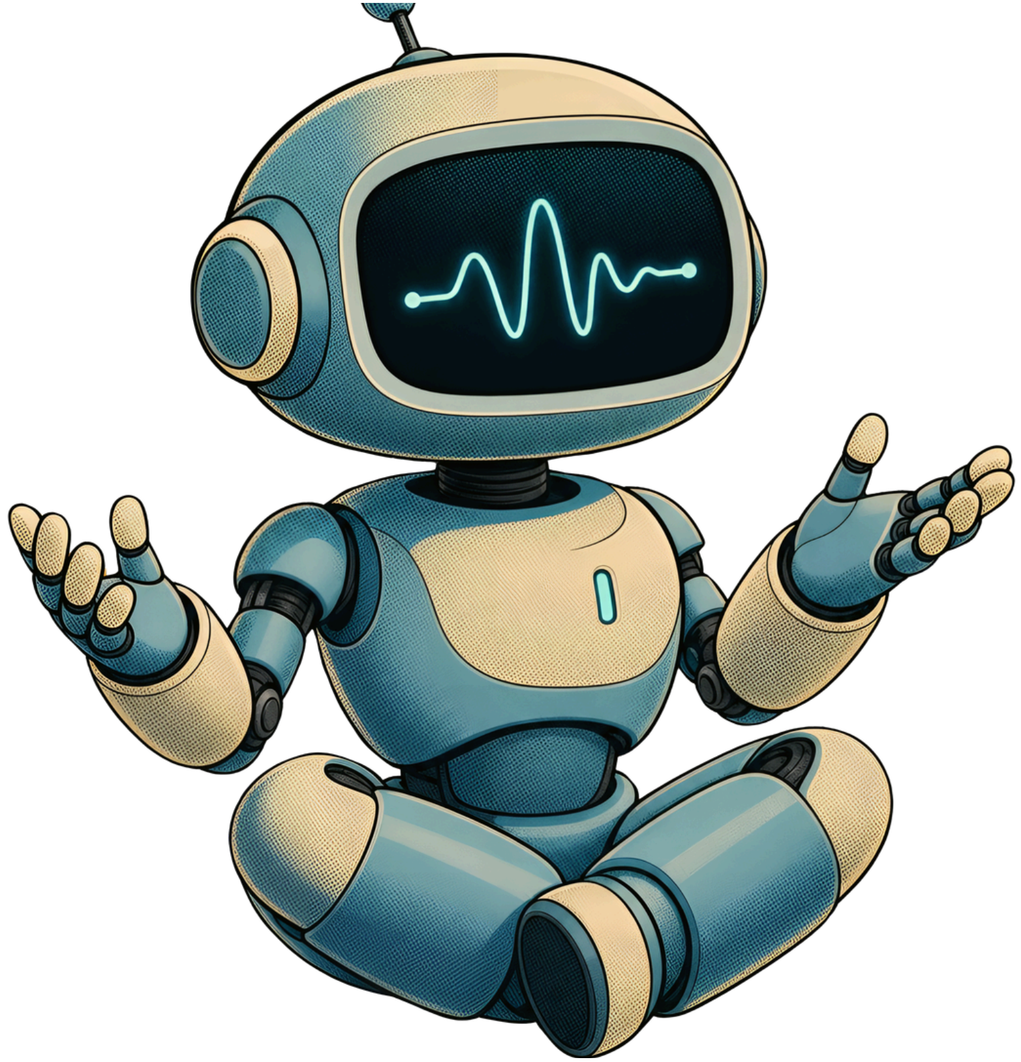


PROBLEM STATEMENT

The problem
Official PDFs are often long, mixed-format, and difficult to search quickly. Some are digital; others contain scanned pages that normal text extraction cannot read.

The user need
A user should be able to upload one document, understand its contents, ask natural-language questions, and receive answers tied back to source pages.

The project goal
Build a web-based assistant that combines native PDF extraction, OCR, semantic retrieval, grounded LLM generation,summaries, key points and action items.



TOOLS AND TECHNOLOGIES

BACKEND	FASTAPI + UVICORN
TEXT READING	PYMUPDF + TESSERACT OCR
SEARCH	MINILM EMBEDDINGS + CHROMADB
AI MODEL	GOOGLE GEMINI (GOOGLE-GENAI)
USER INTERFACE	GRADIO
SETTINGS	PYDANTIC SETTINGS, .ENV



SYSTEM ARCHITECTURE

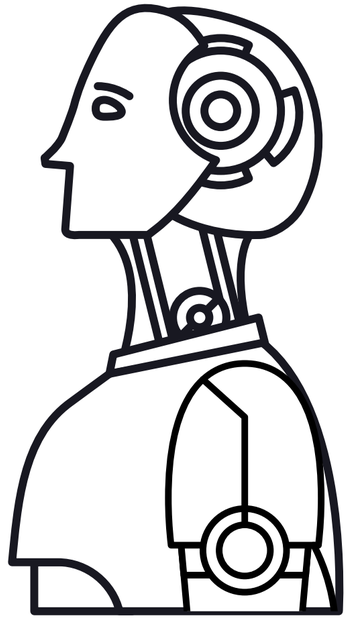


1 UPLOAD CHECK PDF AND SIZE → 2 EXTRACT TEXT, OCR IF SCANNED → 3 SPLIT SMALL CHUNKS WITH PAGES

4 STORE EMBEDDINGS IN CHROMADB → 5 SEARCH FIND BEST CHUNKS → 6 ANSWER GEMINI + SOURCES

GRADIO SCREEN
UPLOAD, ASK AND SUMMARY TABS. IT TALKS TO THE BACKEND OVER HTTP.

FASTAPI BACKEND
ROUTES FOR UPLOAD, CHAT, SUMMARY, KEY POINTS AND ACTION ITEMS.



CORE FEATURES

1.PDF UPLOAD

TYPE AND SIZE CHECKS

2.TEXT AND OCR

SCANNED PAGES HANDLED

3. Q/A

ASK QUESTION RELATED TO DOCUMENT

4.SOURCE REFERENCES

FILE AND PAGE SHOWN

5.SUMMARY

SHORT, CLEAR OVERVIEW

6.KEY POINTS

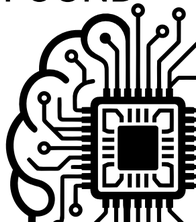
6 TO 10 MAIN POINTS

7.ACTION ITEMS

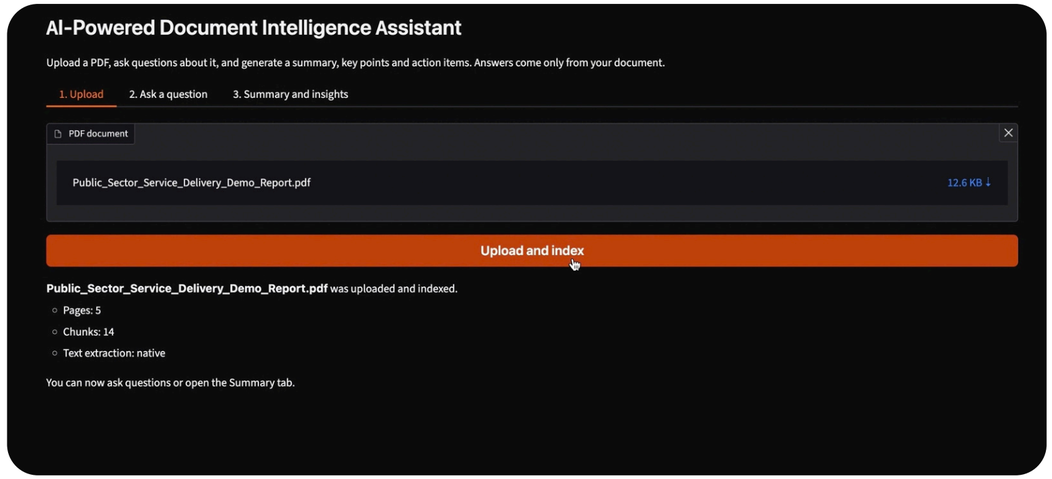
NEXT STEPS FROM THE TEXT

8.FALL BACK RESPONSE

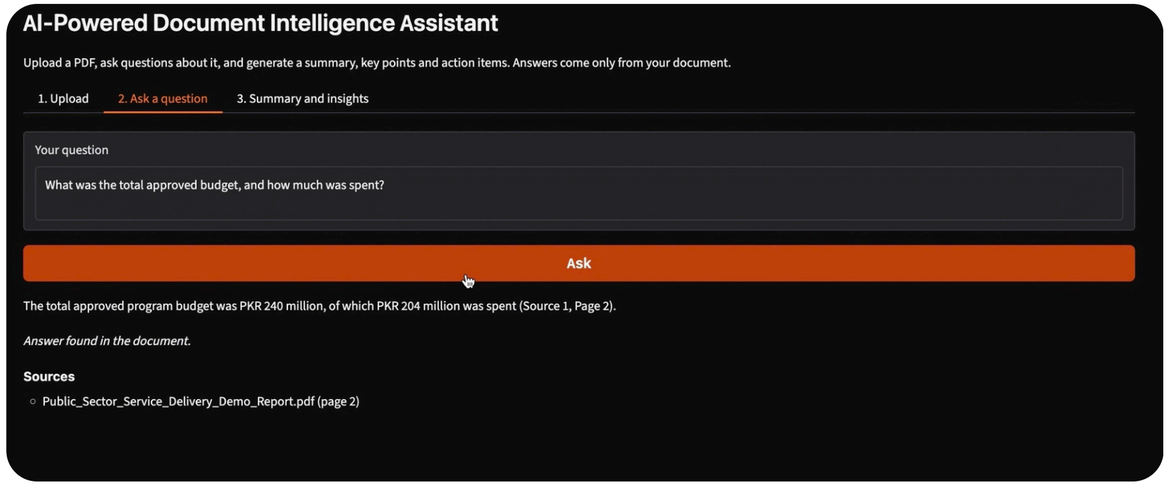
SAYS WHEN IT IS NOT FOUND



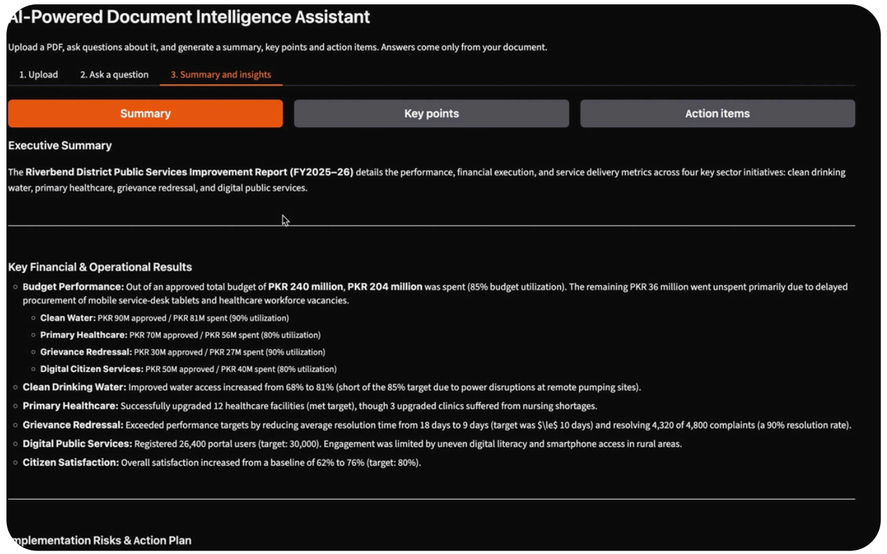
DEMO SCREENSHOTS



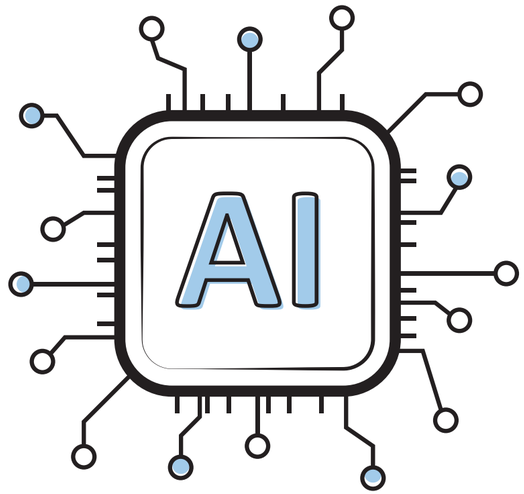
1. UPLOAD



2. ASK A QUESTION



3. SUMMARY AND INSIGHTS



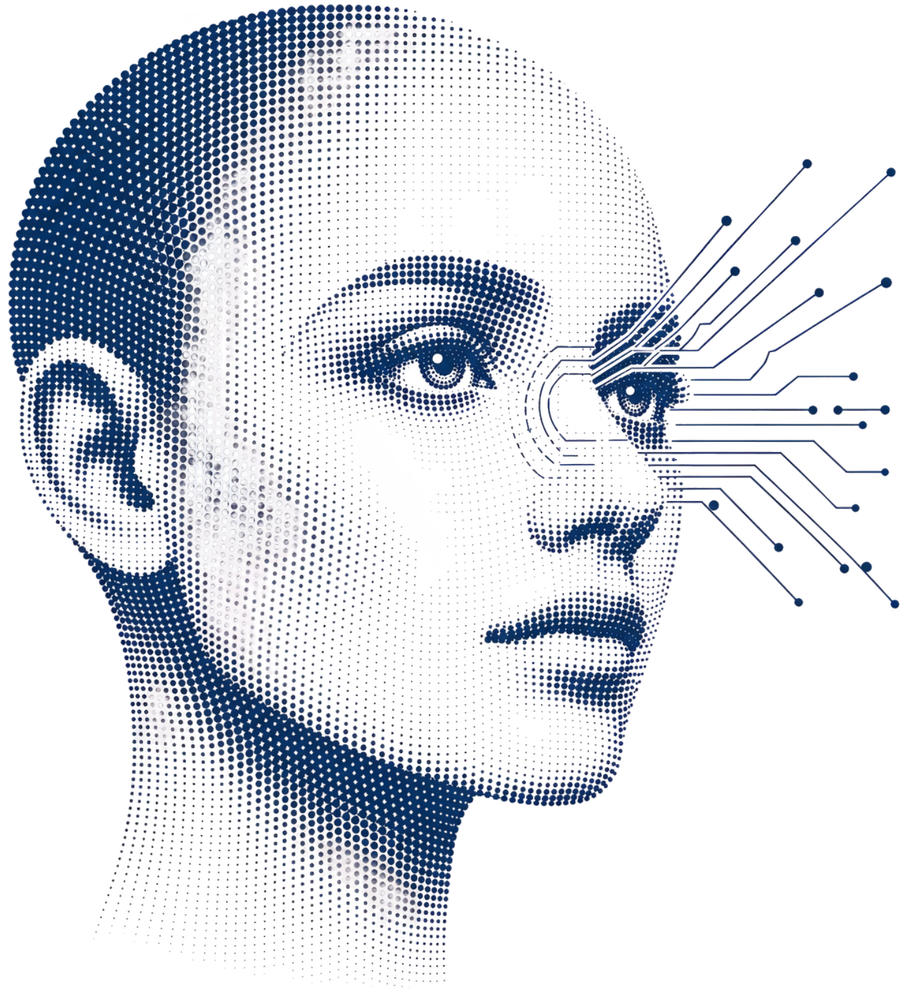
CHALLENGES FACED

Scanned and mixed PDFs
Read native text first. Run OCR only on pages with very little text.

Wrong or missing search results
Tuned chunk size, top 15 results and a similarity limit. Tested with a real budget report.

FAKE OR WRONG CITATIONS
Checked every cited page against the saved page number.

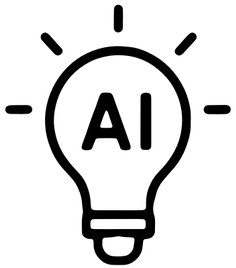
API busy or quota reached
Retry 3 times, then show a clear message to the user.



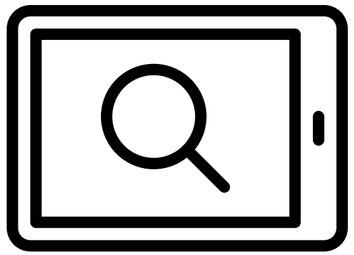
LEARNING OUTCOMES

KEY SKILLS & TECHNICAL LEARNING

- RETRIEVAL-AUGMENTED GENERATION (RAG): BUILT A COMPLETE DOCUMENT-GROUNDED QUESTION-ANSWERING PIPELINE USING GEMINI AND CHROMADB.
- SEMANTIC SEARCH & EMBEDDINGS: LEARNED HOW SENTENCE TRANSFORMERS AND COSINE SIMILARITY RETRIEVE RELEVANT DOCUMENT CHUNKS.
- PDF PROCESSING & OCR: IMPLEMENTED PYMUPDF AND TESSERACT TO EXTRACT TEXT FROM DIGITAL AND SCANNED PDFS.
- PROMPT ENGINEERING: DESIGNED GROUNDED PROMPTS, SOURCE CITATIONS, AND FALLBACK RESPONSES TO REDUCE UNSUPPORTED ANSWERS.
- DEBUGGING & INTEGRATION: IMPROVED RETRIEVAL USING TOP-15 SEARCH, TESTED REAL PUBLIC-SECTOR REPORTS, AND CONNECTED GRADIO WITH FASTAPI USING DOCUMENT IDS.



FUTURE SCOPE & IMPROVEMENTS

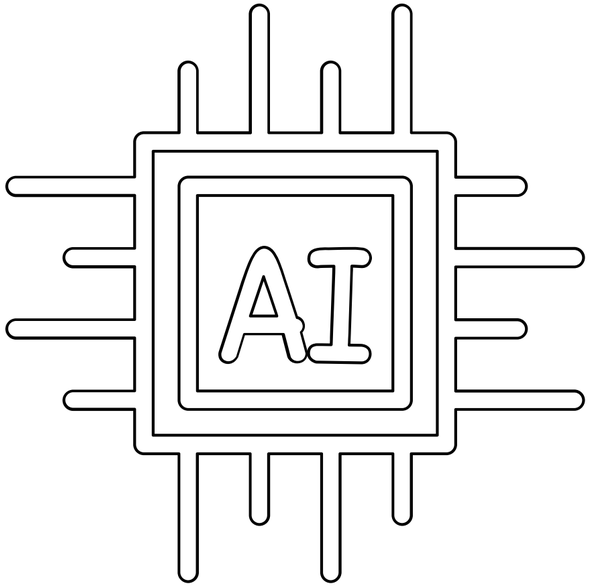


- **Hybrid Search:** Add keyword search (BM25) alongside the existing semantic search for better retrieval.
- **Large PDF Support**: Improve summaries of lengthy documents by processing them section by section.
- **User Security**: Add login, user accounts, and secure document access.
- **Cloud Deployment:** Make the application available online with better performance and document management.
- **Urdu and more languages** : Add Urdu OCR for local public records.
- **Multi-document workspace**: Keep multiple PDFs and filter by active document or collection



CONCLUSION

SUCCESSFULLY DEVELOPED AN AI-POWERED DOCUMENT INTELLIGENCE ASSISTANT THAT SIMPLIFIES PUBLIC-SECTOR DOCUMENT ANALYSIS THROUGH:
PDF UPLOAD WITH NATIVE TEXT EXTRACTION AND OCR SUPPORT.
DOCUMENT-GROUNDED QUESTION ANSWERING WITH SOURCE REFERENCES.
AUTOMATED SUMMARIES, KEY POINTS, AND ACTION ITEM EXTRACTION.
MODULAR FASTAPI BACKEND AND INTERACTIVE GRADIO INTERFACE.
PROJECT IMPACT: DEMONSTRATES HOW AI CAN SUPPORT FASTER INFORMATION RETRIEVAL, REPORT ANALYSIS, AND EVIDENCE-BASED DECISION-MAKING IN PUBLIC-SECTOR WORKFLOWS.
CURRENT LIMITATIONS: SINGLE-DOCUMENT DESIGN, CONTEXT-LIMITED ANALYSIS, AND RETRIEVAL/CITATION ACCURACY THAT CAN BE FURTHER IMPROVED.



GITHUB REPOSITORY:
HTTPS://GITHUB.COM/ABEERZAFF/AI-INTERNSHIP-PITB
THANK YOU!
QUESTIONS & DISCUSSION